Sensor Fusion
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Sensor fusion is the process of combining information from multiple, heterogeneous sensors to produce a unified, more accurate, and more reliable estimate of the state of a system than any single sensor could provide alone. In the context of autonomous driving, this state is typically a vehicle's position, velocity, orientation, and acceleration as it navigates a dynamic environment.

No single sensor is sufficient for safe autonomous operation:

- GPS/GNSS provides absolute position but is unreliable in urban canyons and degrades under multi-path interference.
- LiDAR (Light Detection and Ranging) provides high-resolution three-dimensional point clouds but is expensive and computationally demanding.
- RADAR (Radio Detection And Ranging) performs robustly in adverse weather but has lower spatial resolution.
- IMU (Inertial Measurement Unit) provides high-frequency acceleration and angular rate measurements but drifts (accumulates error) over time.
- Cameras provide rich semantic information but are sensitive to lighting conditions.

Sensor Fusion Problem

Let $\mathbf{x}_t \in \mathbb{R}^n$ denote the true state of the ego-vehicle (the autonomous vehicle) at time t. Let $\{\mathbf{z}_t^{(s)}\}_{s=1}^{S}$ denote the collection of S noisy measurements from S distinct sensors. The fundamental challenge is to compute:

$$p\left(\mathbf{x}_t \,\middle|\, \mathbf{z}_{1:t}^{(1)}, \mathbf{z}_{1:t}^{(2)}, \ldots, \mathbf{z}_{1:t}^{(S)}\right)$$

— the posterior probability distribution of the state given all sensor observations up to and including time t. Under the linear Gaussian assumption, this posterior is exactly Gaussian and the Kalman filter computes it in closed form via a recursive algorithm of $O(n^3)$ complexity per time step.

Sensor Modalities in Autonomous Driving

Global Navigation Satellite Systems (GNSS, including GPS, GLONASS, Galileo) triangulate position by measuring the time-of-flight of signals from multiple satellites. A standard GPS receiver outputs:

- Horizontal position: latitude \phi, longitude \lambda, or equivalent Cartesian coordinates (x, y) in a local East-North-Up (ENU) frame.
- Vertical position: altitude z.
- Update rate: typically 1–10 Hz.
- Accuracy: 1–5 m (consumer grade), 1–10 cm with RTK (Real-Time Kinematic) correction.
- Noise model: approximately Gaussian, $\sigma_{\mathrm{GPS}} \approx 1\text{–}5m$, but with non-Gaussian tails under multi-path.

An IMU combines accelerometers (measuring specific force, i.e., acceleration minus gravity) and gyroscopes (measuring angular rate). It provides:

- Angular rates: $\dot{\phi}$ (roll rate), $\dot{\theta}$ (pitch rate), $\dot{\psi}$ (yaw rate) — collectively denoted $\boldsymbol{\omega}$.

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- Specific force (acceleration): $\mathbf{a}$ in the body frame.
- Update rate: 100–1000 Hz.
- Noise model: white noise plus slowly varying bias $\mathbf{b}$, which must be estimated as part of the state:

$$\tilde{\mathbf{a}} = \mathbf{a}_{\text{true}} + \mathbf{b}_a + \boldsymbol{\eta}_a, \qquad \boldsymbol{\eta}_a \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_a)$$

The IMU bias $\mathbf{b}_a$ is itself modeled as a random walk:

$$\dot{\mathbf{b}}_a = \boldsymbol{\nu}_a, \qquad \boldsymbol{\nu}_a \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_b)$$

A LiDAR sensor emits laser pulses and measures their time of return, constructing a dense 3D point cloud of the environment at 10–20 Hz. For localization purposes, a scan matching algorithm (*e.g.*, ICP — Iterative Closest Point) compares successive point clouds or matches against a pre-built HD map to yield a pose estimate $(\hat{x}, \hat{y}, \hat{\psi})$ with covariance $\mathbf{R}_{\text{LiDAR}}$.

RADAR emits radio waves and measures the range r and range rate $\dot{r}$ (radial velocity via the Doppler effect) of reflecting objects. It provides:

- Robust operation in rain, fog, and darkness.
- Direct measurement of relative velocity (a key advantage over GPS and LiDAR).
- Lower angular resolution than LiDAR.
- Update rate: 10–20 Hz.

The table below summarizes how sensor modalities complement each other — a principal motivation for fusion:

Property	GPS	IMU	LiDAR	RADAR	Camera
Absolute position	✓	✗	✓ (map)	✗	✗
Velocity	✗	✓ (integrated)	✗	✓ (Doppler)	✗
Works in rain/fog	✓	✓	✗	✓	✗
Works indoors/tunnels	✗	✓	✓	✓	✓
Long-term stable	✓	✗ (drift)	✓	✓	✗
High frequency	✗	✓	✗	✗	✗

<u>State Space Formulation for Vehicle Dynamics</u>

For a vehicle operating in a 2D plane (sufficient for many highway and urban scenarios), a common state vector is the kinematic bicycle model state:

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$$\mathbf{x}_t = \begin{pmatrix} x_t \\ y_t \\ \psi_t \\ v_t \\ \dot{\psi}_t \\ a_t \end{pmatrix}$$

where:

- (x_t, y_t): position in a local Cartesian frame (metres)
- ψ_t: heading (yaw angle, radians) — the direction the vehicle faces
- v_t: longitudinal speed (m/s)
- $\dot{\psi}_t$: yaw rate (rad/s)
- a_t: longitudinal acceleration (m/s²)

For full 3D navigation (e.g., for drones or off-road vehicles), the state is extended to include altitude, pitch, roll, and their rates — a 15-dimensional state is standard in INS/GNSS integration (T*itterton & Weston*, 2004).

The simplest motion model assumes constant velocity with additive Gaussian process noise. In a 2D *Cartesian* frame with state $\mathbf{x}_t = (x_t,\ y_t,\ \dot{x}_t,\ \dot{y}_t)^\top$:

$$\mathbf{x}_{t+1} = \mathbf{F}\mathbf{x}_t + \mathbf{G}\mathbf{w}_t$$

$$\mathbf{F} = \begin{pmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{G} = \begin{pmatrix} \frac{\Delta t^2}{2} & 0 \\ 0 & \frac{\Delta t^2}{2} \\ \Delta t & 0 \\ 0 & \Delta t \end{pmatrix}, \quad \mathbf{w}_t \sim \mathcal{N}\big(\mathbf{0},\ \sigma_a^2 \mathbf{I}_2\big)$$

Here Δt is the time step and σ_a^2 is the process noise variance representing un-modeled accelerations. The process noise covariance is:

$$\mathbf{Q} = \mathbf{G}\,\sigma_a^2 \mathbf{I}_2\,\mathbf{G}^\top = \sigma_a^2 \begin{pmatrix} \frac{\Delta t^4}{4} & 0 & \frac{\Delta t^3}{2} & 0 \\ 0 & \frac{\Delta t^4}{4} & 0 & \frac{\Delta t^3}{2} \\ \frac{\Delta t^3}{2} & 0 & \Delta t^2 & 0 \\ 0 & \frac{\Delta t^3}{2} & 0 & \Delta t^2 \end{pmatrix}$$

The Constant Turn Rate and Velocity (CTRV) model is the standard choice for road vehicles turning at approximately constant yaw rate. The continuous-time dynamics are:

$$\dot{\mathbf{x}} = \begin{pmatrix} v\cos\psi \\ v\sin\psi \\ \dot{\psi} \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ \nu_a \\ \nu_{\dot{\psi}} \end{pmatrix}$$

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with state $\mathbf{x} = (x, y, \psi, v, \dot{\psi})^\top$. After *Euler* integration over Δt (assuming constant $\dot{\psi}$ during the interval), the discrete-time transition is:

$$\mathbf{x}_{t+1} = f(\mathbf{x}_t) + \mathbf{w}_t$$

where the non-linear function f is:

$$f(\mathbf{x}_t) = \begin{cases} \begin{pmatrix} x_t + \frac{v_t}{\dot{\psi}_t}\left[\sin(\psi_t + \dot{\psi}_t \Delta t) - \sin\psi_t\right] \\ y_t + \frac{v_t}{\dot{\psi}_t}\left[-\cos(\psi_t + \dot{\psi}_t \Delta t) + \cos\psi_t\right] \\ \psi_t + \dot{\psi}_t \Delta t \\ v_t \\ \dot{\psi}_t \end{pmatrix} & \text{if } |\dot{\psi}_t| > \epsilon \\ \begin{pmatrix} x_t + v_t \Delta t \cos\psi_t \\ y_t + v_t \Delta t \sin\psi_t \\ \psi_t \\ v_t \\ \dot{\psi}_t \end{pmatrix} & \text{if } |\dot{\psi}_t| \leq \epsilon \end{cases}$$

The degeneracy at $\dot{\psi}_t = 0$ (straight-line motion) is handled by the second branch, evaluated when $|\dot{\psi}_t| < \epsilon$ (*e.g.*, $\epsilon = 10^{-4} rad/s$). The non-linearity of f necessitates the EKF or UKF.

<u>Fundamental Assumptions</u>

Assumption 1: Markov Property of Vehicle Dynamics
The future state $\mathbf{x}_{t+1}$ depends only on the current state $\mathbf{x}_t$ and the current control input, not on the full history $\mathbf{x}_{1:t-1}$:

$$p(\mathbf{x}_{t+1} \mid \mathbf{x}_t, \mathbf{x}_{t-1}, \ldots) = p(\mathbf{x}_{t+1} \mid \mathbf{x}_t)$$

Justification: *Newton*'s laws of motion are *Markovian* — given full knowledge of position, velocity, and acceleration, the future trajectory is determined by the equations of motion. The state vector must be chosen to be sufficiently comprehensive to satisfy this property. If the state vector is too small (*e.g.*, position only), the Markov property is violated because velocity is hidden.

Assumption 2: White Process Noise
The process noise $\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ is independent across time steps.
Justification: Un-modeled forces (wind gusts, road irregularities, driver micro-corrections) are, to first approximation, uncorrelated at the time scales of interest. Serial correlation in the residuals is a diagnostic signal that the process model is misspecified.

Assumption 3: White Measurement Noise
Each sensor's noise $\mathbf{v}_t^{(s)} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}^{(s)})$ is serially uncorrelated and independent across sensors.
Justification: Most sensor noise arises from independent physical sources (thermal noise, quantization, atmospheric fluctuation). Correlated noise from shared systematic errors (*e.g.*,

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ionospheric delay affecting multiple GPS antennas) is a deviation from this assumption and must be modeled explicitly.

Assumption 4: Gaussianity of Noise Distributions
Both process and measurement noise are Gaussian.
Justification: By the central limit theorem, noise arising from many small, independent contributions is approximately Gaussian. For a standard Kalman filter, Gaussianity is required for optimality; for the EKF and UKF, it is used as an approximation. Particle filters relax this assumption entirely at the cost of increased computation.

Assumption 5: Known and Time-Invariant Noise Covariances
The matrices $\mathbf{Q}$ and $\mathbf{R}^{(s)}$ are assumed known.
Justification and caveat: In practice, these covariances are estimated from manufacturer specifications, empirical calibration, or adaptive filtering schemes (e.g., the innovation-based covariance estimation of Mohamed & Schwarz, 1999). Mis-specification of $\mathbf{R}$ is a major source of filter divergence in deployed systems.

Assumption 6: Sensor Independence Conditional on the State
Given the true state $\mathbf{x}_t$, the measurements from different sensors are independent:

$$p(\mathbf{z}_t^{(1)}, \ldots, \mathbf{z}_t^{(S)} \mid \mathbf{x}_t) = \prod_{s=1}^{S} p(\mathbf{z}_t^{(s)} \mid \mathbf{x}_t)$$

Justification: Physical processes such as GPS multi-path and LiDAR reflection arise from independent physical phenomena. Violations occur when sensors share a field of view and observe the same physical artifact (e.g., a retroreflective surface saturating both a camera and LiDAR). Such common-mode errors must be guarded against in system design.

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